Smart Helmet: What We Can Build on Our Own Servers

From the supplier's Interface Document. This is a plan and nothing is built yet.

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01 What runs on our own servers

Once a helmet is told our address it talks only to us. These are the parts we run in South Africa. The supplier's servers are not needed after that.

Part	What it does	When
Address server	Tells each helmet where to report. One small web page (pp3 and 4).	v1.00
Live server	Keeps the live connection with every helmet. Takes the login and the reports and the alarms and sends commands back (pp10 to 21).	v1.00
Data store	Keeps every report and alarm so the control room can show history.	v1.00
Control room app	Our own screens for the control room, replacing the supplier screens.	v1.00
File server	Receives the photos and audio and video that helmets upload (pp22 to 34).	v3.00
Video server	Receives live video from a helmet (p37).	v4.00
Call server	Lets the control room talk to a helmet on a video call (p39).	v4.00

ONE THING TO ADD OURSELVES

The helmet logs in with its device number only and there is no password (p10). Our live server will keep a list of allowed device numbers and refuse the rest.

02 What the document lets us build

Each line is something the helmet does and what the document says about it. Page numbers are from the Interface Document.

What the helmet does	What the document gives us	Can we build it?
1 Find our server	The helmet asks an address and gets two addresses back. The reply also holds its settings such as alarm temperature and how long before a removed helmet alarms (pp3 to 6).	Yes
2 Log in	The helmet sends its device number and we reply with its details (pp10 to 14). The login uses only the device number so our server must check a list of allowed numbers.	Yes
3 Report position and status every 10 seconds	Position and accuracy and battery and whether it is worn and 4G or Wi-Fi and signal and SIM and memory card and temperature (pp15 to 19). Positions use the Gaode map system (question 3).	Yes
4 Alarms	SOS and fall and helmet off and silent and temperature and fence and near electricity and altitude (p20). The SOS button sends its own message with a call room number (p21).	Yes
5 Bluetooth beacons	Every report lists the Bluetooth devices the helmet hears with signal strength and time (p15). A setting turns scanning on (p6). Alarms exist for entering and leaving a beacon (p20). Three fields in our reply are not explained: f_id and f_key and bt_beacon (p19).	Partly. We can read and keep the list. How the helmet picks beacons is not explained.
6 Data saved with no signal	The document names ca_backup_data as the offline upload and does not describe it (p2).	Not yet. We need the format.
7 Fences	get_user_fence and the fence status fields are named but not described (pp2 and 18).	Not yet. We need the format.

03 What the document lets us build, continued

What the helmet does	What the document gives us	Can we build it?
8 Commands to the helmet	Start and stop recording and take a photo (p27). Ask for nearby Wi-Fi and tell it to join one (pp41 to 43). Play audio (p25). Change the volume (p11).	Yes
9 Photos and audio and video files	The helmet uploads to our address by HTTP. We can ask for the list of videos it holds and choose which to send (pp22 to 34).	Yes. It needs our file server.
10 Live video	We send a start command with an address to stream to (p37).	Later. It needs our video server.
11 Video calls	The helmet uses a standard call system so it needs a call server (p39).	Later. It needs our call server.
12 Software updates	Our reply to a helmet can hold a new software address and a check number (pp5 and 8). Each helmet asks for itself (p3) so we can decide for each one.	Yes to send. How a helmet accepts it is questions 8 and 9.

04 The order we would build it in

Version	What we build	What it needs
v1.00: surface	Address server and live server and data store. Screens for the overview and one worker and alarms and setup. A Bluetooth test list.	Nothing. We can start. A real helmet needs its admin login.
v2.00: underground	Beacon names and the trail with times. The offline upload.	Answers to questions 4 and 5.
v3.00: control and files	Commands and photos and audio and video and the Wi-Fi setting.	Our file server.
v4.00: live video and calls	Live video and video calls.	Our video server and call server.
Updates	Sending software updates to chosen helmets.	Answers to questions 8 and 9.